HACK MATRIX HACKATHON

TITLE PAGE

- **PROJECT TITLE :** **BioDex - Gamified Urban Wildlife Tracker**
  Under Wildlife Sighting Reporter **Problem ID:** HM44
- **TEAM NAME :** **Debug Duo**
- **TEAM MEMBERS :**

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PROBLEM STATEMENT

- **Describe the problem clearly .**

  Urban biodiversity data is incredibly scarce because everyday citizens lack a simple, structured, and engaging way to contribute local wildlife observations.

- **Who is affected by this problem .**

  Environmental researchers tracking urban ecosystems, and local communities disconnected from their surrounding wildlife.

- **Why this problem is important ?**

  Without crowdsourced data, it is impossible to monitor species decline, invasive species spread, or the general ecological health of our immediate urban environments.

EXISTING SOLUTIONS AND GAPS

- **List of existing Solutions (if any)**

  Complex academic portals (like eBird) and unstructured posts in local community WhatsApp/Facebook groups.

- **Mention their Limitations**

  Academic tools have steep learning curves that deter casual users. Social media posts lack standardized GPS tagging and searchable databases.

- **Explain the gap your project will address**

  BioDex bridges the gap by turning ecological data collection into an engaging, gamified experience for the everyday citizen, automatically capturing standard GPS and timestamp data without extra effort.

PROPOSED SOLUTION

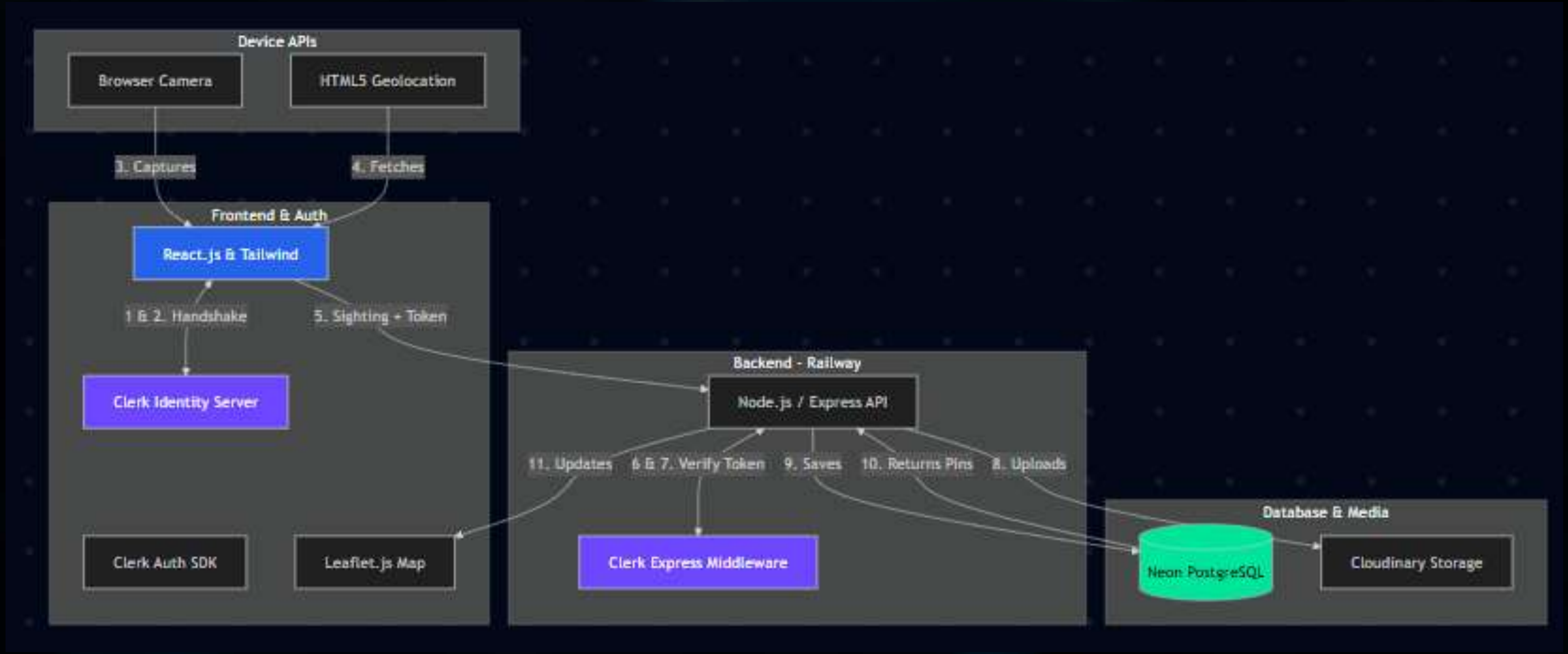
- **Explain your solution clearly .** A mobile-responsive web app where users photograph local wildlife, log it with automatic GPS coordinates, and view community sightings on an interactive map.

- **List key features** A photo-first sighting submission form, an interactive real-time community map, and a gamified "Species Explorer" badge system, secure user authentication.

- **Explain how it solves the problem** It removes the friction of data entry and introduces a "collect-them-all" motivation, seamlessly generating rich, localized biodiversity data.

SYSTEM ARCHITECTURE

- **Include architecture diagram**



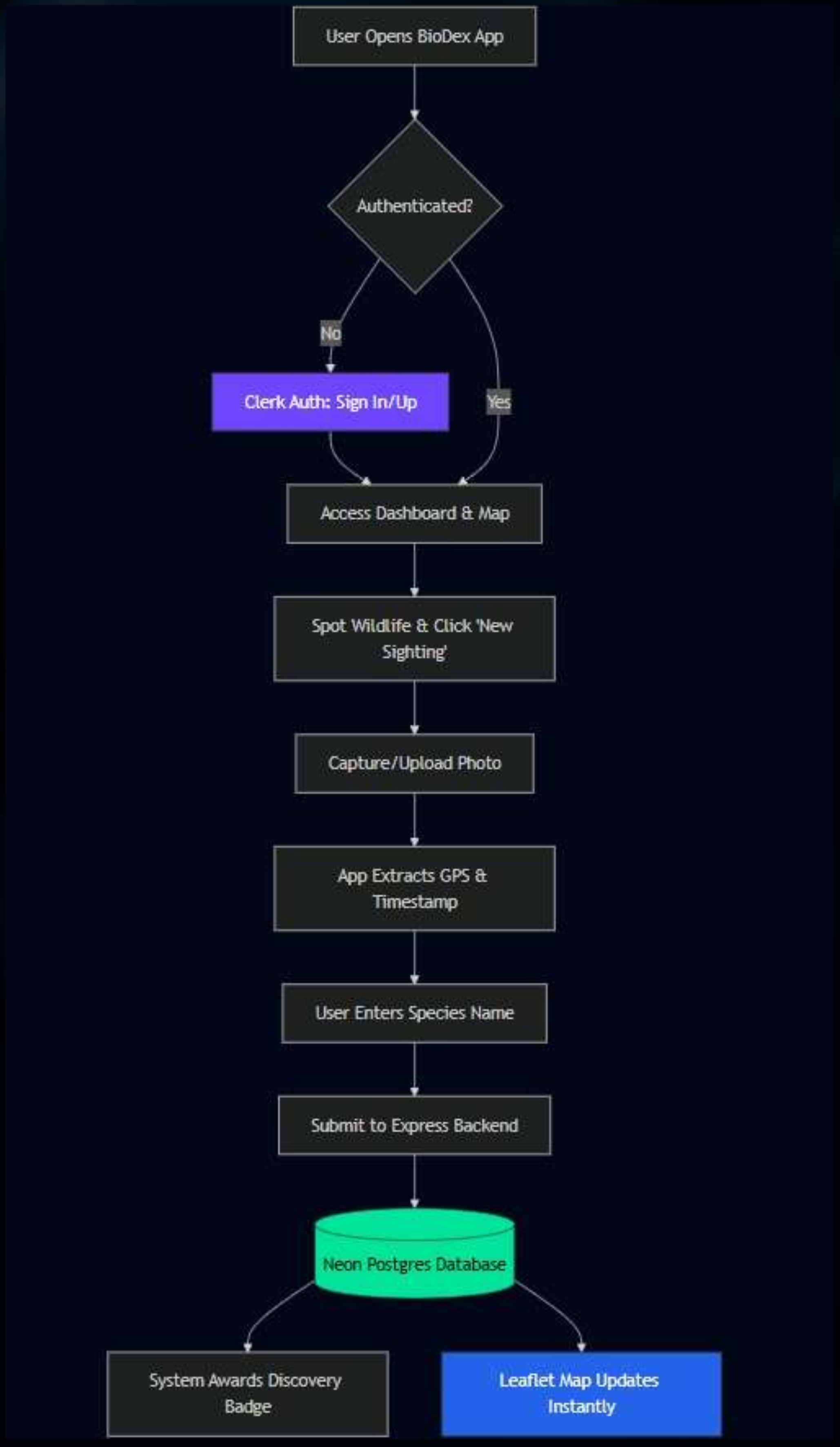
- **Mention main components (Frontend, Backend, Database, APIs)**
  - Frontend: React.js & Clerk Auth SDK (Handles User Sign-in & Session Management)
  - Backend: Node.js / Express.js with Clerk Middleware
  - Database: PostgreSQL via Neon Tech (Relational Data) & Cloudinary (Image Storage)
  - APIs: HTML5 Geolocation API, Leaflet.js , Gemini API

- **Explain data flow briefly**

  The React client captures location and photo -> Photo goes to Cloudinary -> Backend receives Image URL + GPS data -> Saves to Neon Postgres Database -> Pushes data back to update the Leaflet map.

FLOWCHART / WORKING PROCESS

- **Show step-by-step working**
- **Include user flow or system flow**
- **Use a proper flowchart**



TECH STACK

- **Frontend technologies** — React (Vite) and Tailwind CSS - For rapid, responsive mobile-first UI development.

- **Backend technologies** — Node.js with Express (Hosted on Railway) - For robust REST API creation and fast CI/CD deployment.

- **Database** — Neon Tech (Serverless PostgreSQL) for sighting data, and Cloudinary for wildlife image storage.

- **Tools and APIs used and reason for choosing each (1 line each)**
  - Leaflet.js: Lightweight, open-source mapping without requiring API keys.
  - Clerk Authentication,Gemini API
  - HTML5 Geolocation API: Native browser support to easily capture accurate coordinates.

IMPLEMENTATION PLAN

- **Step-by-step development plan**

1) Scaffold UI and initialize Map.
2) Build photo upload and GPS capture logic.
3) Configure Clerk Auth & Middleware
4) Render dynamic pins on the map.

- **Modules or features to be built**

Interactive Map View, Sighting Submission Form, Gamification/Badge Engine.

- **Timeline (if applicable)**

- 9:30 AM - 12:30 PM: Core React UI, Map setup, and Clerk Provider Integration.
- 12:30 PM - 3:30 PM: Form logic & Geolocation API.
- 3:30 PM - 5:30 PM: Styling and Testing

EXPECTED OUTCOME / IMPACT

- **Expected results of the project**

A functional prototype featuring an interactive map populated with wildlife pins and a working submission form capturing real location data.

- **Benefits to users**

Citizens get a fun, engaging way to interact with nature, learn about local species, and earn digital rewards.

- **Real-world impact**

Creates a valuable, crowdsourced dataset of urban biodiversity that can be exported for environmental research and local conservation planning.

FUTURE SCOPE AND CONCLUSION

- **Future improvements**

  Integrating an AI Vision model to automatically identify the animal in the uploaded photo, reducing user entry error.

- **Final summary**

  Expanding to a nationwide platform with regional leaderboards and partnerships with schools for environmental education.

- **Scalability of the project**

  BioDex proves that by combining accessible mapping tech with gamified user experiences, we can solve the urban biodiversity data shortage one photo at a time. Expanding Clerk to include multiple social logins (GitHub, Apple) for frictionless onboarding.